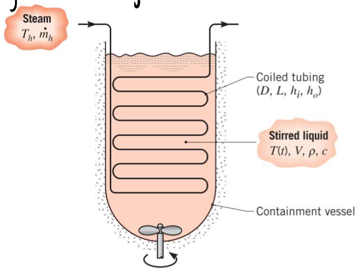
Homework 5

Saturday, February 15, 2020 5:42 PM

100

5.10) $\rho = 1200\ kg/m^3 \quad c = 2200\ J/kg\cdot K \quad V = 2.25\ m^3$



$T_i = 300K \rightarrow T = 450K$

$T_h = 500K$

$D = 20\ mm = .02\ m$

$h_i = 10{,}000\ W/m^2 K$

$h_o = 2000\ W/m^2 \cdot K$

$t = 60\ min = 3600s$

L?

$$U = \left(\frac{1}{h_i} + \frac{1}{h_o}\right)^{-1} = \left(\frac{1}{10000} + \frac{1}{2000}\right)^{-1}$$

$$\frac{dU}{dt} = \rho Vc \frac{dT}{dt} = -UA_s(T - T_h)$$

$$\int_{T_i}^{T} \frac{dT}{T - T_h} = \frac{-UA_s}{\rho Vc}\int_0^t dt \rightarrow \ln\left(\frac{T - T_h}{T_i - T_h}\right) = \frac{-UA_s t}{\rho Vc}$$

$$A_s = \frac{\rho Vc}{Ut} \ln\left(\frac{T_i - T_h}{T - T_h}\right) = \frac{(1200)(2.25)(2200)}{(1666.7)(3600)} \ln\left(\frac{300 - 500}{450 - 500}\right)$$

$$A_s = 1.37\ m^2 = \pi D L$$

$$L = \frac{A_s}{\pi D} = \frac{1.37}{\pi(.02)} \rightarrow \boxed{L = 21.8m}$$

28

5.58) $D = 100\ mm = .1\ m$ (sphere) $T_i = 50°C \quad T_\infty = 20°C$

$k = 2 W/mK \quad \rho = 2500\ kg/m^3 \quad C_p = 750\ J/kg K \quad h = 30\ W/m^2 K$

@ $T = 26°C$ t? E?

$$\frac{\rho Vc}{hA_s} \ln\left(\frac{T_i - T_\infty}{T - T_\infty}\right) = t = \frac{(2500)(\frac{4}{3})\cancel{\pi}(.05)^{\cancel{3}}(750)}{(30)\cancel{(4)}\cancel{\pi}\cancel{(.05)^2}} \ln\left[\frac{50 - 20}{26 - 20}\right]$$

$\boxed{t = 1676.5\ sec}$

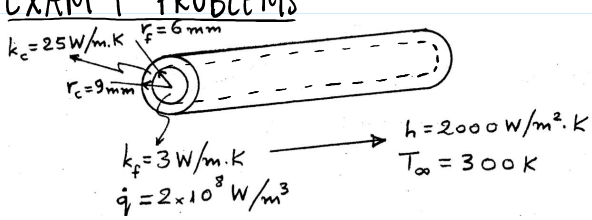
$$E = hAt(T_i - T) = 30(4)\pi\left(\tfrac{.05}{3}\right)^2(1676.5)(50 - 26)$$

$\boxed{E = 4213.7J}$

28

EXAM 1 PROBLEMS

1)



a. Equations for $T_f(r)$ & $T_c(r)$

FROM $0 \le r \le r_f \rightarrow$ ① $\frac{1}{r}\frac{\partial}{\partial r}\left(k_f r \frac{\partial T_f}{\partial r}\right) + \dot{q} = 0$ 2 BC

FROM $r_f \le r \le r_c \rightarrow$ ② $\frac{1}{r}\frac{\partial}{\partial r}\left(k_c r \frac{\partial T_c}{\partial r}\right) = 0$ (NO HEAT GEN) 2 BC

① $\int \partial\left(k_f r \frac{\partial T_f}{\partial r}\right) = \int -\dot{q}r\ \partial r$

$k_f r \frac{\partial T_f}{\partial r} = -\frac{\dot{q}r^2}{2} + C_1$

$$k_f r \frac{\partial T_f}{\partial r} = -\frac{\dot{q} r^2}{2} + C_1$$

BC 1: $\left.\frac{\partial T_f}{\partial r}\right|_{r=0} = 0 \rightarrow C_1 = 0$ $\rightarrow \int \partial T_f = \int -\frac{\dot{q} r}{2k_f}\, \partial r$

ⓐ $T_f = -\frac{\dot{q} r^2}{4k_f} + C_2$

BC 2: @ $r = r_f$ $T_c = T_f$ $\rightarrow$ ⓑ $T_f(r_f) = -\frac{\dot{q} r_f^2}{4k_f} + C_2 = T_c$

② $\int \frac{\partial}{\partial r}\left(k_c r \frac{\partial T_c}{\partial r}\right) = \int 0 \rightarrow k_c r \frac{\partial T_c}{\partial r} = C_3$

BC 3: $\left. k_c \frac{\partial T_c}{\partial r}\right|_{r=r_f} = \left. k_f \frac{\partial T_f}{\partial r}\right|_{r=r_f} \longrightarrow k_c \frac{\partial T_c}{\partial r} = \frac{C_3}{r_f} = \frac{\dot{q} r_f}{2} \rightarrow C_3 = -\frac{\dot{q} r_f^2}{2}$

$\int \partial T_c = \int \frac{C_3}{k_c r}\, \partial r \rightarrow$ ⓒ $T_c = \frac{C_3}{k_c} \ln(r) + C_4 = -\frac{\dot{q} r_f^2}{2k_c} \ln(r) + C_4$

BC 4: $\left. -k_c \frac{\partial T_c}{\partial r}\right|_{r=r_c} = h\left[T_c(r_c) - T_\infty\right] \rightarrow \frac{\dot{q} r_f^2}{2r_c} = h\left(-\frac{\dot{q} r_f^2}{2k_c}\ln(r_c) + C_4 - T_\infty\right) \rightarrow C_4 = \frac{\dot{q} r_f^2}{2r_c h} + \frac{\dot{q} r_f^2}{2k_c}\ln(r_c) + T_\infty$

FROM ⓑ, ⓒ, & solved C_3 & C_4:

$$-\frac{\dot{q} r_f^2}{4k_f} + C_2 = -\frac{\dot{q} r_f^2}{2k_c}\ln(r_f) + \frac{\dot{q} r_f^2}{2r_c h} + \frac{\dot{q} r_f^2}{2k_c}\ln(r_c) + T_\infty$$

$$C_2 = \frac{\dot{q} r_f^2}{4k_f} + \frac{\dot{q} r_f^2}{2k_c}\ln\left(\frac{r_c}{r_f}\right) + \frac{\dot{q} r_f^2}{2r_c h} + T_\infty$$

FINALLY:

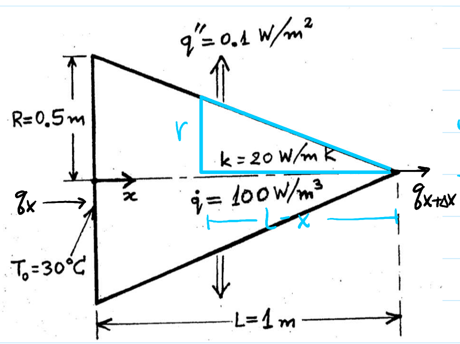
$$\boxed{\begin{aligned} T_f(r) &= \frac{\dot{q}}{4k_f}(r_f^2 - r^2) + \frac{\dot{q} r_f^2}{2k_c}\ln\left(\frac{r_c}{r_f}\right) + \frac{\dot{q} r_f^2}{2r_c h} + T_\infty \\ T_c(r) &= \frac{\dot{q} r_f^2}{2r_c h} + \frac{\dot{q} r_f^2}{2k_c}\ln\left(\frac{r_c}{r}\right) + T_\infty \end{aligned}}$$

b. MAX T_f IN FUEL ELEMENT $\rightarrow T_f(r=0)$?

$$T_f(0) = \frac{(2\times10^8)(.006)^2}{4(3)} + \frac{(2\times10^8)(.006)^2}{2(25)}\ln\left(\frac{.009}{.003}\right) + \frac{(2\times10^8)(.006)^2}{2(.009)(2000)}$$

$\boxed{T_f = 1158\text{ K}}$

2)



SIMILAR TRIANGLES:

$\frac{r}{R} = \frac{L-x}{L}$

$r = R\left(1 - \frac{x}{L}\right)$

ENERGY BALANCE: (CONSIDER CYLINDER)

IN - OUT + GEN - CONS = $\cancel{ACC}^{0}$

$q_x - q_{x+\Delta x} + \dot{q}V - q_{cond} = 0$

$q_x - \left(q_x + \frac{d}{dx} q_x\, dx\right) + \dot{q}\pi r^2 dx - 2\pi r q'' dx = 0$

$$\text{IN} - \text{OUT} + \text{GEN} - \text{CONS} = \cancel{\text{ACC}}$$

$$q_x - q_{x+\Delta x} + \dot{q}V - q_{cond} = 0$$

$$q_x - \left(q_x + \frac{d}{dx} q_x \, dx\right) + \dot{q}\pi r^2 dx - 2\pi r q'' dx = 0$$

$$-\frac{d}{dx}\left(-kA_s \frac{dT}{dx}\right) dx + \dot{q}\pi r^2 dx - 2\pi r q'' dx = 0$$

$$-k\frac{d}{dx}\left(A_s \frac{dT}{dx}\right) \cancel{dx} = \dot{q}\pi r^2 \cancel{dx} - 2\pi r q'' \cancel{dx}$$

$$\int d\left(\pi r^2 \frac{dT}{dx}\right) = \int \frac{-\dot{q}\pi R^2\left(1-\frac{x}{L}\right)^2}{k} + \frac{2\pi q'' R\left(1-\frac{x}{L}\right)}{k} \, dx \qquad u = 1-\frac{x}{L} \qquad du = -\frac{1}{L}dx$$

$$\pi R^2\left(1-\frac{x}{L}\right)^2 \frac{dT}{dx} = \frac{\dot{q}\cancel{\pi R^2} L\left(1-\frac{x}{L}\right)^3}{3k} - \frac{\cancel{2\pi} q'' R L \cancel{\left(1-\frac{x}{L}\right)^2}}{\cancel{2}k} + C_1$$

$$\frac{dT}{dx} = \frac{\dot{q}L\left(1-\frac{x}{L}\right)}{3k} - \frac{q''L}{Rk} + \frac{C_1}{\pi R^2\left(1-\frac{x}{L}\right)^2}$$

$$\underline{\text{BC 1}}: \left.\frac{dT}{dx}\right|_{x=L} = 0 \ (\text{INSULATED}) \rightarrow 0 = 0 - \frac{q''L}{RK} + \frac{C_1}{0} \quad \therefore C_1 = 0$$

$$\int dT = \int \frac{\dot{q}L\left(1-\frac{x}{L}\right)}{3k} - \frac{q''L}{RK} \, dx \qquad u = 1-\frac{x}{L} \qquad du = -\frac{1}{L}dx$$

$$T(x) = \frac{-\dot{q}L^2\left(1-\frac{x}{L}\right)^2}{6k} - \frac{q''Lx}{RK} + C_2$$

$$\underline{\text{BC 2}}: T(x=0) = T_0 = 30°C \rightarrow T_0 = \frac{-\dot{q}L^2}{6k} + C_2 \rightarrow C_2 = T_0 + \frac{\dot{q}L^2}{6k}$$

$$\boxed{T(x) = \frac{-\dot{q}L^2}{6k}\left(1-\frac{x}{L}\right)^2 - \frac{q''Lx}{RK} + \frac{\dot{q}L^2}{6k} + T_0}$$

$$T(L) = \frac{-(0.1)(1)(1)}{(0.5)(20)} + \frac{(100)(1)^2}{6(20)} + 30 \longrightarrow \boxed{T(L) = 30.83°C}$$